Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details



Wai Lin (Leon)
Team lead: product and code



Aung Nyein Chan (Michael)
Also Lin-Chiang Lee
Architect: BIM model, field validation

a. Team name: TrueSite
b. Team leader name: Wai Lin (Leon)
c. Problem Statement: Future of Work & Enterprise Productivity

Site crews build without the BIM model in front of them,
so mistakes surface weeks later as rework.

Brief about the idea

AR × BIM × AI for construction crews

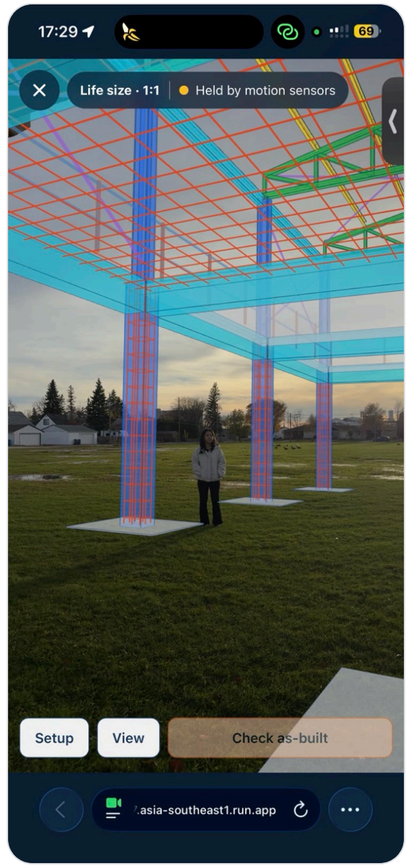
Site crews build without the BIM model in front of them, so mistakes surface weeks later as rework. TrueSite puts the model, and a second pair of eyes, where the work happens.

1. **Scan**
   A printed QR code, fixed at a surveyed spot, opens TrueSite in the phone's browser.
2. **See**
   The BIM model appears over the real site at true scale. No app, no headset.
3. **Check**
   Tap **Check as-built**: Gemini on Google Cloud compares what is built with the design, pins the deviations on the model and drafts the issue.

14+ h a week per project team member lost to hunting for project data, resolving conflicts and rework

48% of all rework on US jobsites comes from miscommunication and poor project data

Source: FMI & PlanGrid, "Construction Disconnected" (2018), a survey of nearly 600 construction leaders



Live AR at life size on an iPhone, field test, 17 Oct 2026, on the deployed service

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Different from existing tools

- **AR viewers** on the market (Dalux, Trimble, GAMMA AR, XYZ Reality) need an app, a headset or survey hardware, and leave the comparison to the person looking.
- **AI progress tools** need capture rigs and work after the fact.
- **TrueSite** takes the camera's pose from the QR code, renders the design from that viewpoint, and has Gemini judge each element against its own outline.

How it solves the problem

- **The model on site:** scan a printed code and the design stands at 1:1 over the site.
- **Answers on the spot:** tap any element for its BIM data, or ask Gemini about it.
- **Caught early:** Check as-built finds a deviation while it can still be fixed, not weeks later as rework.
- **Paperwork done:** Gemini drafts the issue in the crew's language; a person saves it.

Unique selling point

- **One flow:** AR + BIM + AI from a printed code.
- **No app, no headset:** any phone browser.
- **Grounded:** element-level checks against the BIM data.
- **Safe by design:** severity set by rule in code, never by the model.
- **For every crew:** issues in their own language, one of eight.

List of features offered by the solution

1. **Live AR at life size**
   The model at 1:1 over the real site, from a printed QR code.
2. **Desk sheet at 1:100**
   The whole building on a printed sheet, for a desk or a meeting.
3. **Tap for BIM data**
   Any element's type, size, material and properties, from the model.
4. **Ask Gemini**
   Answers come only from the element's own data, citing the fields.
5. **Check as-built**
   Three aligned images; findings pinned, severity set by fixed rules.
6. **Findings in 3D**
   Design ghosts show each deviation; picking a finding turns the view to it.
7. **Issues drafted by Gemini**
   In one of eight crew languages; a person edits, saves and copies it.
8. **Daily site report**
   Gemini writes the day's report from the checks, safety first.
9. **3D model viewer**
   Orbit, plan view, X-ray and a colour key, with Ask and the last check.
10. **View controls**
    Transparency, a section cut, view distance and layers, on site.
11. **Virtual demo**
    Four QR spots and ten planted defects; each check is scored against them.
12. **Any device, no app**
    A phone, tablet or laptop browser; a laptop hands live AR to a phone.

Process flow diagram or Use-case diagram

See it

On site, in the phone's browser

1. **Scan the QR code**
   Printed at a surveyed spot
2. **Calibrate once and lock**
   Tilt for 10 s; the code's corners give the pose
3. **See the model at life size**
   1:1 over the site, held by the motion sensors

Check it

With Gemini on Google Cloud

4. **Tap or ask**
   Any element's BIM data; Ask Gemini about it
5. **Check as-built**
   Gemini compares the photo with the design from the same pose
6. **Findings pinned**
   On the model, with "Why it matters" and a severity set by rule

Act on it

People decide

7. **Draft the issue**
   Gemini drafts it in the crew's language
8. **A person saves it**
   Gemini's draft becomes an issue only when a person saves it
9. **Site report**
   Gemini writes the day's report, safety first

Guardrails | "Visual check, not a survey." | Severity comes from code, never from the model. | Gemini has no tools and no write access.

Wireframes/Mock diagrams of the proposed solution (optional)



1 · Scan

The phone's camera reads the printed code and opens TrueSite



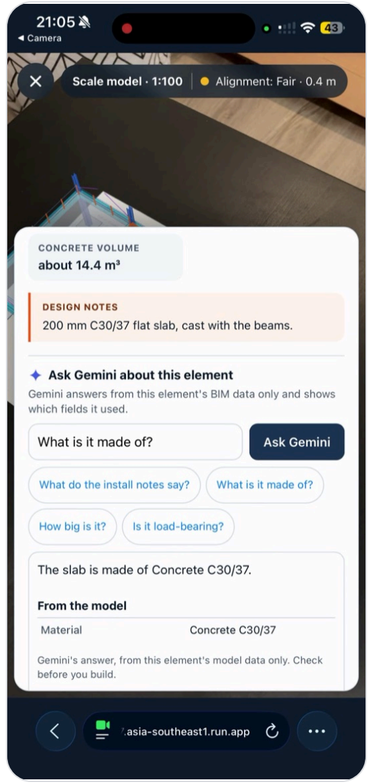
2 · Calibrate and lock

Tilt for 10 s; the code's corners give the camera pose



3 · See and tap

The design at 1:1; tap column B1 for its size and concrete grade



4 · Ask Gemini

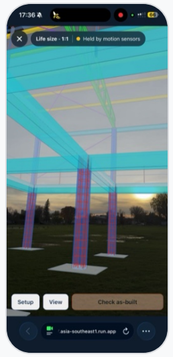
Gemini answers from the element's own data, citing the field

Real screens from our iPhone field test, 17 Oct 2026, on the deployed Cloud Run service (step 4 on the 1:100 desk sheet)

Architecture diagram of the proposed solution

Phone browser

React + three.js, one web app



- QR reading: zxing-wasm in a web worker
- Pose from the code's corners
- Sensor fusion, gravity levelling
- The model drawn at true scale
- ID-buffer render: numbered marks, design render, outlines

3 images + element list

findings + severity

Google Cloud **Also:** Cloud Build · Artifact Registry · Secret Manager · Budget alerts

Cloud Run

asia-southeast1
The FastAPI API and the web app on one origin

- Limits and daily caps
- Schema repair
- Severity by rule

Gemini 3.8 Flash

On Gemini Enterprise Agent Platform, global
Busy or hung: 3.5 Flash, then 3.5 Flash-Lite

Cloud Firestore

Through the Firebase Admin SDK
Check runs, issues, the daily cap

Cloud Storage

Photos and renders
Deleted after 30 days

Gemini has no tools and no write access; its JSON is schema-checked.

Technologies to be used in the solution

Google Cloud and Firebase	Cloud Run, asia-southeast1 · Cloud Firestore · Firebase Admin SDK · Cloud Storage · Cloud Build · Artifact Registry · Secret Manager · Budget alerts
AI	Gemini 3.8 Flash · Gemini Enterprise Agent Platform · Gemini 3.5 Flash and 3.5 Flash-Lite as fallbacks · Google Gen AI SDK · Structured JSON output
Web, AR and API	React · three.js · TypeScript · Vite · zxing-wasm in a web worker · Motion sensors · FastAPI on Python 3.13
BIM	IFC, the open ISO 16739 standard · IfcOpenShell · IFC → glTF, each element keyed by its IFC GlobalId · Printable QR sheets
Testing	Vitest · pytest · Playwright · axe-core · StrykerJS · cosmic-ray · Hypothesis · Schemathesis

Estimated implementation cost (optional)

US$0.015

per as-built check

Measured on the deployed service: about 6,900 input and 2,600 output tokens

US$6.60

a month for a heavy user

Gemini at 20 checks a day for 22 days, about US$0.015 each

Scales to zero

Cloud Run, when idle

At most 2 instances on the deployed service

US$19

per field seat a month

Proposed. Viewing is free for anyone who scans a code.

Limits that cap the bill

- 30 checks per IP per hour, and 300 a day across all users
- Questions, issue drafts and site reports share a daily budget of 300
- 4 MB per uploaded image
- Photos and renders deleted after 30 days
- Budget alerts on the Google Cloud project

Free to view, priced per field seat

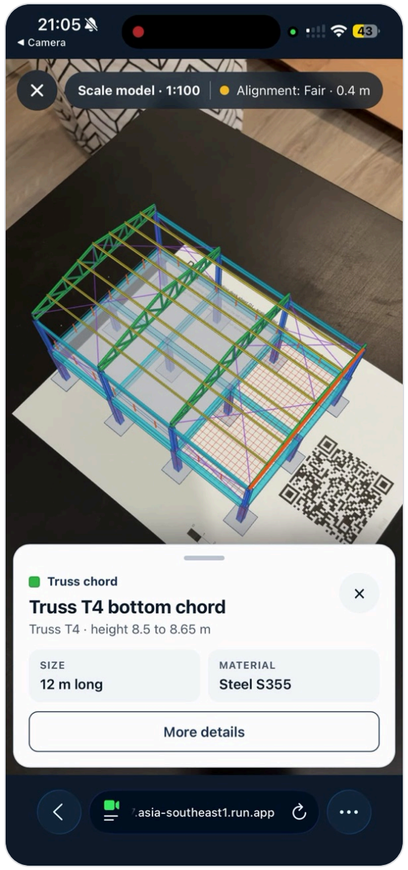
- Anyone on site can scan a code and view the model, so subcontractors join for free.
- Contractors pay per field seat for checks, Ask and issues: US$19 a month, to validate in the pilot.
- A heavy user's checks cost about US$6.60 a month in Gemini, well under the seat price.

Proposed pricing, to validate in the pilot. Costs come from the usage logged on our deployed service.

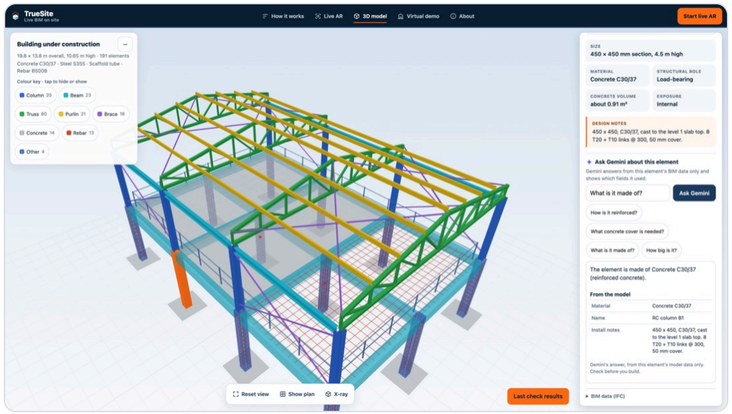
Snapshots of the prototype



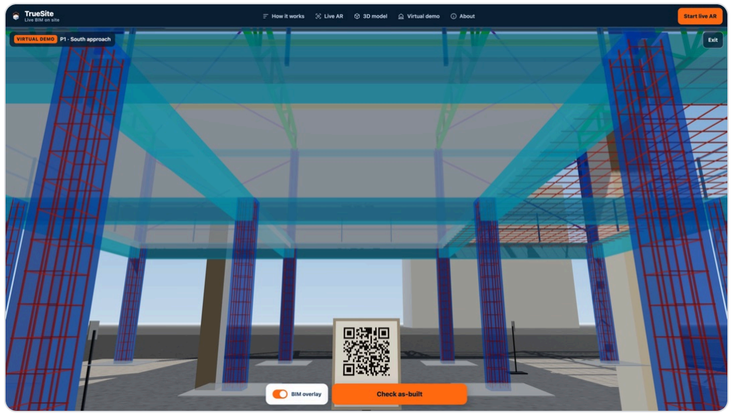
Live AR at 1:1 on a field: a guardrail tapped



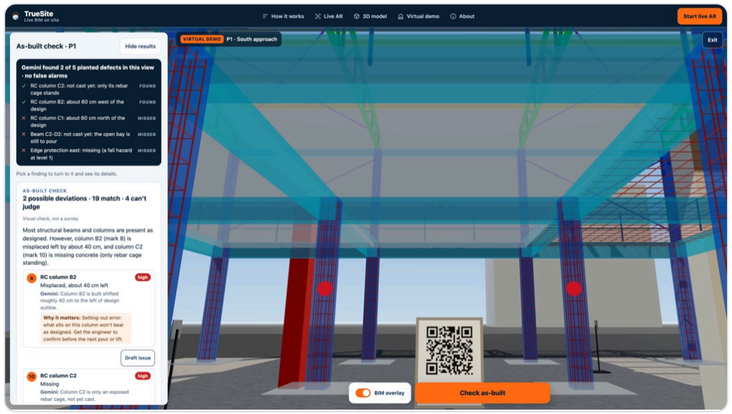
Desk sheet at 1:100: a truss chord tapped



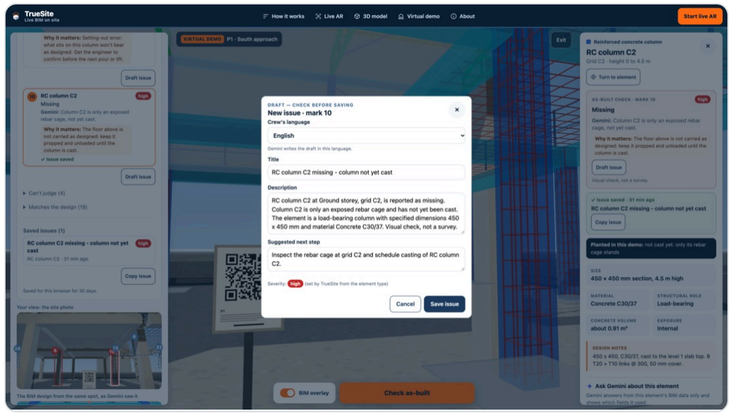
3D model viewer with Ask Gemini



Virtual demo at P1, BIM overlay on



Check results: scorecard and findings



Gemini's draft issue, for a person to save

Prototype Performance report/Benchmarking

Measurement	Result	Source
Pose accuracy, 0.5 px corner noise, code at 0.5–3 m	**median 3.6 mm and 0.66°**	Simulation
Self-calibration from 30 tilted views	**focal length within 1%**	Simulation
Steadiness on a stand, jitter a frame	**under 0.6 px, down from 1–44 px**	Ray-traced test videos
Virtual demo check at P1, 5 planted defects in view	**found 2, no false alarms**	Deployed service
Checks at P1–P4, 16 planted defects in view, 3 live runs	**found 12, 11 and 7; false alarms 5, 8 and 0**	Simulated site, live Gemini
Same checks with two images, no outlines	**found 8; false alarms 5 (one run)**	Simulated site, live Gemini

Automated tests, 18 Oct

Web
549
passed

Vitest

Server
529
passed, 9 skipped

pytest

BIM pipeline
88
passed

pytest

End to end
164
passed, 4 skipped

Playwright

Still to validate on a device

- Overlay error on a real phone at 1–3 m
- Checks scored on real site photos
- The 18 Oct steadiness fix on the iPad

Simulation numbers are not field accuracy. We publish what we measure.

Additional Details/Future Development (if any)

BIM + AI: where construction is going

- Singapore requires BIM submissions for buildings over 5,000 m² (BCA). The UK has required BIM on centrally procured public projects since April 2016.
- Construction is one of the least digitised industries: second to last in the US and last in Europe on McKinsey's index.
- The next step: take the model to the crew, and let AI read it.

Works with the BIM you already have

- IFC, the open format that Revit, Archicad, Tekla and most BIM tools export, read by the open-source IfcOpenShell.
- Prepared once: a glTF model with each element keyed by its IFC GlobalId, plus the element data and the QR sheets.
- No vendor lock-in: the model stays in open IFC, and the crew needs only a browser.

Roadmap

Now
Prototype live on Cloud Run, field-tested on an iPhone on 17 Oct 2026.

→

Next: an 8-week pilot
One or two Singapore contractors, 10 field seats, with labelled real-site photos to measure the check.

→

Then: Singapore, then JAPAC
S$47–53 billion of construction demand projected for Singapore in 2026 (BCA). Taiwan and the rest of JAPAC next.

Product next	World-locked AR on Android and walk-inside on iPhone: built behind a flag, not yet tested on a device	BCF export and connectors for common data environments	Multi-code site setup

Provide links to your:

- GitHub Public Repository https://github.com/wl-geek/truesite
- Demo Video Link (3 Minutes) https://youtu.be/M8zh02kMtk4 3:15 · also on the start page
- Final Product Link https://truesite-1067044308557.asia-southeast1.run.app

Try it in a few minutes, no printer needed

1. Open the product link and tap **Try the virtual demo**.
2. Step in at a QR spot, P1 to P4, and look around.
3. Tap **Check as-built**: Gemini compares your view with the design.
4. Tap **Draft issue** on a deviation, edit it and save it.
5. Tap **Write today's site report**: Gemini sums up the checks, safety first.
6. Open the 3D model, tap any element and **Ask Gemini** about it.

With a printer: download the desk sheet, print it at 100% and scan it for live AR at 1:100.

Scan to open TrueSite
on a phone. Nothing to install.

Business case and go-to-market

Who pays

US$19

per field seat a month, proposed

Viewing is free for anyone who scans a code. Contractors pay per field seat for checks, Ask and issues, at a price to validate in the pilot.

Cost to serve

US$0.015

per check, measured

A heavy user at 20 checks a day for 22 days costs about US$6.60 a month in Gemini. Cloud Run scales to zero.

Singapore first

S$47–53 billion

projected 2026 construction demand

BIM submission is already mandatory over 5,000 m². Then Taiwan and the rest of JAPAC.

An 8-week pilot

10 field seats

with one or two Singapore contractors

Success: deviations caught before pouring, time to answer a model question on site, false-alarm rate.

Less rework. Faster sign-off. Safer sites.

Catch it before it's poured

A missing cage or a shifted column found on site, while it can still be fixed.

Safer for workers

Guardrail and edge-protection deviations always rate high, and issues reach the crew in their own language.

Anyone on the crew

A phone and a printed code: no app, no headset. Subcontractors join by scanning.

Sources: Building and Construction Authority, Singapore (Jan 2026 outlook; BIM e-submission). Check cost logged on our deployed service.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

